

MACHRIHANISH CAMPBELTOWN, United Kingdom

WMO: 031110

Lat: 55.4408N

Lon: 5.6969W

Elev: 33

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	27.4	30.3	21.9	16.5	31.5	24.5	18.6	33.7	42.8	43.2	39.6	42.9	4.9	100	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	69.4	61.5	66.3	60.1	64.0	58.9	63.0	67.1	61.4	64.8	60.0	62.8	12.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.2	80.9	64.6	59.8	76.9	62.9	58.6	73.6	61.5	28.5	67.2	27.4	65.2	26.4	63.0	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.0	31.9	28.6	22.4	75.0	3.7	4.0	19.7	77.9	17.6	80.2	15.5	82.4	12.8	85.3		
(5)			21.4	66.2	3.6	2.2	18.7	67.8	16.6	69.1	14.6	70.4	12.0	72.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	42.0	41.8	43.5	46.2	50.5	54.7	57.6	57.8	55.5	51.0	46.1	42.7		
	DBStd	7.33	4.68	4.79	4.32	4.23	4.32	3.67	3.37	3.22	3.49	4.40	5.01	5.81		
	HDD50	1266	250	230	205	127	43	3	0	0	3	41	132	232		
	HDD65	5789	714	649	667	563	450	310	232	224	286	434	567	693		
	CDD50	959	1	2	3	13	59	144	236	243	167	72	15	4		
	CDD65	6	0	0	0	0	0	1	3	2	0	0	0	0		
	CDH74	13	0	0	0	0	1	1	8	3	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	13.9	17.1	16.4	15.1	13.6	12.4	11.1	10.6	11.2	13.0	14.9	15.6	15.9		
	PrecAvg	51.5	5.4	4.4	3.8	3.3	3.0	3.1	3.3	3.9	3.9	5.9	5.9	5.6		
(15) Precipitation	PrecMax	61.7	10.0	8.4	7.2	5.9	5.6	6.1	5.7	8.6	7.6	11.3	9.3	11.1		
	PrecMin	39.5	0.8	1.4	1.5	1.5	0.6	0.8	0.9	1.3	0.7	1.6	3.2	2.3		
	PrecStd	6.3	2.0	2.0	1.3	1.3	1.4	1.3	1.1	1.8	1.7	2.6	1.8	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	54.9	57.0	63.0	69.8	71.6	74.3	72.8	67.7	62.1	58.4	55.8		
		MCWB	51.8	52.4	52.0	53.9	60.0	60.8	64.6	64.1	62.1	58.2	55.9	53.1		
	2%	DB	52.1	52.0	53.5	58.0	63.9	66.5	69.3	67.8	64.4	59.9	56.2	53.9		
		MCWB	49.9	49.5	49.6	51.7	55.6	59.2	61.9	61.9	59.7	57.0	54.0	51.8		
	5%	DB	50.2	50.2	51.3	54.7	60.3	63.1	65.9	64.9	62.2	58.4	54.5	52.2		
		MCWB	48.4	48.1	48.1	49.8	53.7	57.5	60.1	60.1	58.2	55.8	52.4	50.3		
	10%	DB	48.8	48.7	49.7	52.4	57.2	60.6	63.2	63.0	60.7	57.2	53.2	50.5		
		MCWB	47.0	46.7	47.1	48.5	52.2	56.2	58.8	58.9	57.2	54.7	51.1	48.6		
	0.4%	WB	52.2	52.8	52.6	55.5	60.6	62.9	66.0	65.4	63.4	59.3	56.7	53.8		
		MCDB	53.9	54.3	55.4	60.6	68.5	69.1	72.6	70.6	66.1	61.0	57.8	55.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	50.5	50.1	50.5	52.4	57.0	60.2	62.9	62.9	61.0	57.8	54.5	52.1		
		MCDB	51.8	51.3	52.6	56.5	62.1	64.8	67.3	66.4	63.2	59.3	55.8	53.6		
	5%	WB	48.9	48.5	48.9	50.6	54.9	58.4	61.1	61.3	59.4	56.5	52.9	50.6		
		MCDB	50.1	49.9	50.6	53.7	59.0	62.1	64.7	63.9	61.2	58.0	54.3	52.1		
	10%	WB	47.3	47.1	47.6	49.1	53.1	56.8	59.5	59.8	58.2	55.2	51.2	48.9		
		MCDB	48.7	48.5	49.3	51.8	56.3	60.0	62.4	62.1	60.0	56.9	52.9	50.4		
	5% DB	MDBR	7.9	8.7	10.0	11.0	11.8	10.7	10.3	10.2	9.9	9.2	8.5	8.7		
		MCDBR	8.3	8.3	11.0	13.8	16.3	14.6	15.0	12.7	11.1	9.1	8.2	8.9		
		MCWBR	8.2	7.5	8.4	9.0	9.9	9.0	9.1	8.2	7.8	7.7	8.0	8.6		
		MCWBR	8.1	8.0	9.9	12.0	14.1	13.0	12.8	11.1	9.7	8.4	8.0	8.7		
(35) Clear-Sky Solar Irradiance	taub		0.338	0.349	0.370	0.384	0.387	0.380	0.384	0.368	0.365	0.348	0.330	0.326		
		taud	2.230	2.323	2.320	2.325	2.368	2.421	2.417	2.473	2.476	2.440	2.306	2.197		
	Ebn at Noon		176	227	251	266	272	274	271	269	253	226	181	149		
		Edh at Noon	19	24	31	35	36	34	34	30	26	22	17	16		
(44) All-Sky Solar Radiation	RadAvg		157	375	734	1194	1531	1519	1446	1172	812	440	208	117		
	RadStd		20	41	56	89	139	159	133	87	59	55	21	13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.45	N/A	+1.80	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page